# Pressure & Temperature in Gases

**Exam code:** 0620 & 0971

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**Written by:** Stewart Hird
**Reviewed by:** Lucy Kirkham
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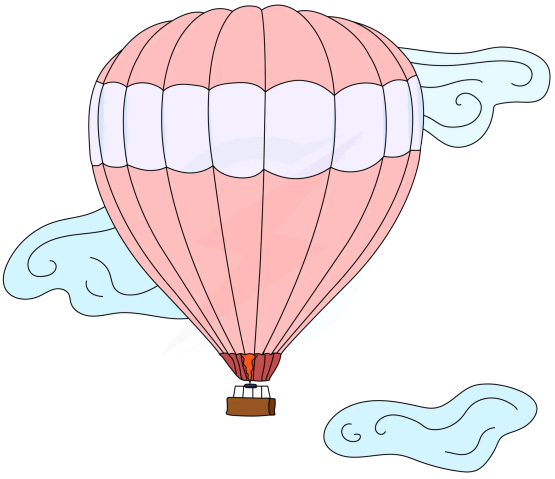
## Pressure & temperature in gases

- A change in temperature or pressure affects the volume of gases
- As the air inside a hot air balloon is heated up, it expands and the balloon gets bigger
- This is because the volume of a gas increases as temperature increases



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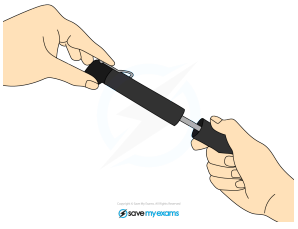
***As temperature increases gas volume increases. The density decreases as the volume increases so the balloon rises.***

- If you have a gas stored inside a container that is squeezed, the pressure increases as you decrease the volume
- This is what happens in a bicycle pump



Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

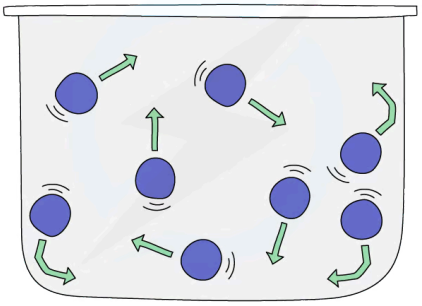
Got it

allows you to inflate a tyre
- You can feel the force of the high pressure if you put your finger on the end of the pump

# Gases & kinetic theory

## Extended tier only

- Gaseous particles are in constant and **random motion**
- The pressure that a gas creates inside a closed container is produced by the gaseous particles hitting the inside walls of the container:



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***Moving particles of gas colliding with each other and the container walls***

## How does temperature affect the volume of a gas?

- Increasing the temperature increases the **kinetic** energy of each particle
  - **Remember:** The thermal energy from increasing the temperature is converted to kinetic energy in the particles
- As the temperature increases, the particles in the gas **move faster** and **spread out more**
- If the gas particles are inside a container, they will collide with the container walls more **frequently**
  - If the container walls are flexible and stretchy then the container will get bigger and bigger, just like the hot air balloon!

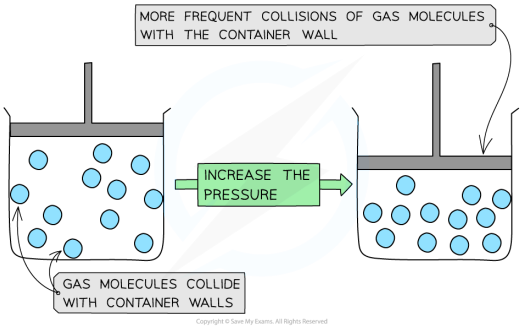
### Examiner Tips and Tricks

If you are talking about the particles, then make sure that you talk about them spreading out.

You will lose a mark in an exam if you talk about particles expanding!

## How does pressure affect the volume of a gas?

- Pressure is about the number of particles in a given volume
- Increasing the pressure means that there are the same number of particles but in a smaller volume
  - Conversely, decreasing the pressure means that there are the same number of particles but in a larger volume




***When the pressure increases, the volume decreases. This means that the molecules collide with the container walls more frequently***

- Since the volume is decreased, the gas particles hit the container walls more frequently
  - If the pressure is too high, this can result in the container leaking gas or exploding

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